

CUSTOMER FEEDBACK SENTIMENT ANALYSIS

CODTECH Internship - Project 3

Technology: Python - Data Science & Analytics

Submitted by

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1. INTRODUCTION

Customer feedback contains valuable information about customer satisfaction, product quality, service experience, and areas that require improvement. Manually analyzing a large number of customer reviews can be time-consuming.

The Customer Feedback Sentiment Analysis project analyzes customer feedback and classifies the feedback into three sentiment categories: Positive, Neutral, and Negative.

The project uses Python for data cleaning, text preprocessing, analysis, visualization, and sentiment prediction. A machine-learning classification model is developed using TF-IDF text features and Logistic Regression. Power BI is used to create a dashboard for presenting sentiment results and customer feedback information.

The project demonstrates an end-to-end data science workflow starting from customer feedback data and ending with sentiment prediction and dashboard visualization.

2. OBJECTIVE

- To analyze customer feedback using Python.
- To clean and preprocess textual customer feedback.
- To classify customer feedback into Positive, Neutral, and Negative sentiments.
- To visualize sentiment distribution.
- To develop a sentiment prediction model.
- To evaluate the prediction model using classification metrics.
- To create a Power BI dashboard.
- To present customer feedback and sentiment predictions clearly.

3. SCOPE

The project focuses on analyzing textual customer feedback and identifying the sentiment expressed in each review. The project covers data preprocessing, sentiment analysis, visualization, machine-learning prediction, model evaluation, and Power BI dashboard creation.

4. TOOLS & TECHNOLOGIES

Technology	Purpose
Python	Data processing, analysis and machine learning
Pandas	Dataset loading and data manipulation
Matplotlib	Data visualization
Scikit-learn	TF-IDF feature extraction and Logistic Regression
Power BI	Interactive dashboard creation
Visual Studio Code	Writing and executing Python code
GitHub	Project repository and submission

5. DATASET DESCRIPTION

The project uses a customer feedback dataset prepared for sentiment analysis. The dataset contains customer review text together with its sentiment label.

The main columns are:

Column	Description
review_id	Unique review identifier
feedback	Customer feedback text
sentiment	Sentiment label

The dataset contains 110 customer feedback records.

The sentiment categories used in the dataset are:

- Positive
- Neutral
- Negative

Dataset usage in the project

The labeled feedback is used for exploratory analysis, visualization, and supervised sentiment classification. A separate test portion is used to evaluate the prediction model.

6. DATA CLEANING AND PREPROCESSING

The customer feedback data was processed before applying the machine-learning model.

- Loaded the customer feedback CSV file using Pandas.
- Converted feedback values into text format.
- Converted text to lowercase.
- Removed unnecessary web links.
- Removed special characters and unwanted symbols.
- Removed extra spaces.
- Prepared cleaned feedback for text feature extraction.
- Checked sentiment labels.
- Divided the dataset into training and testing data.

The cleaned text was then used as input to the TF-IDF vectorizer.

Result: After preprocessing, the dataset was ready for sentiment analysis, visualization, machine-learning prediction, and Power BI dashboard creation.

7. SENTIMENT ANALYSIS

Sentiment analysis is used to identify the emotional classification of customer feedback.

Positive

Feedback expressing satisfaction, appreciation, or a favorable experience.

Neutral

Feedback expressing an ordinary, balanced, or neither clearly positive nor negative experience.

Negative

Feedback expressing dissatisfaction, problems, complaints, or an unfavorable experience.

The sentiment labels are analyzed to understand the overall distribution of customer feedback. The prediction results generated by the model are also examined to determine how the model classifies the test feedback.

8. KEY FINDINGS

The sentiment analysis produced the following observations for the 28 test feedback records:

Measure	Result
Positive predicted feedback	12
Negative predicted feedback	9
Neutral predicted feedback	7
Total test feedback	28
Model accuracy	89.29%

The sentiment distribution demonstrates that all three sentiment categories are represented in the test results.

The classification report also provides precision, recall, and F1-score for each sentiment category. These measures provide a more detailed view of model performance across the three classes.

9. DATA VISUALIZATIONS

The project includes visualizations to make the sentiment analysis results easier to understand.

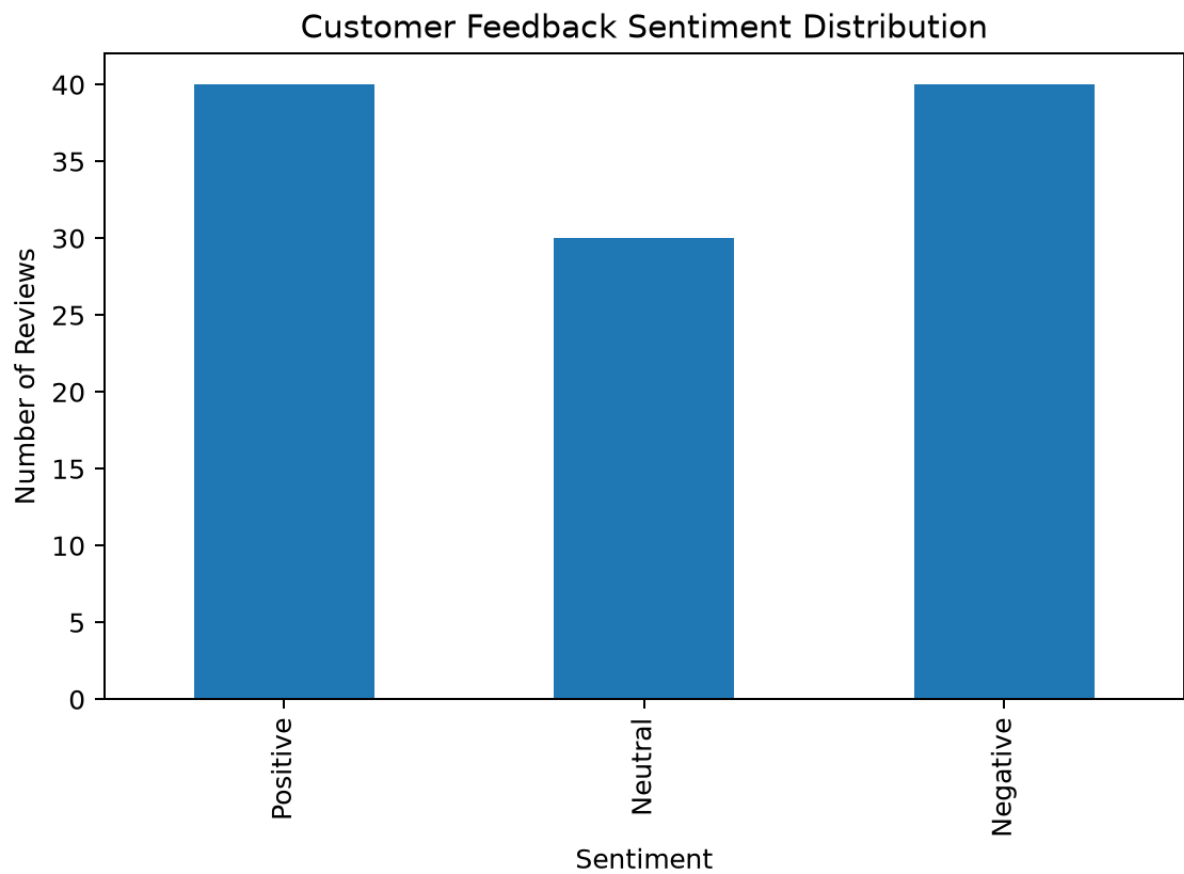


Figure 1: Customer Feedback Sentiment Distribution

The chart presents the number of labeled customer feedback records belonging to each sentiment category.

9. DATA VISUALIZATIONS (CONTINUED)

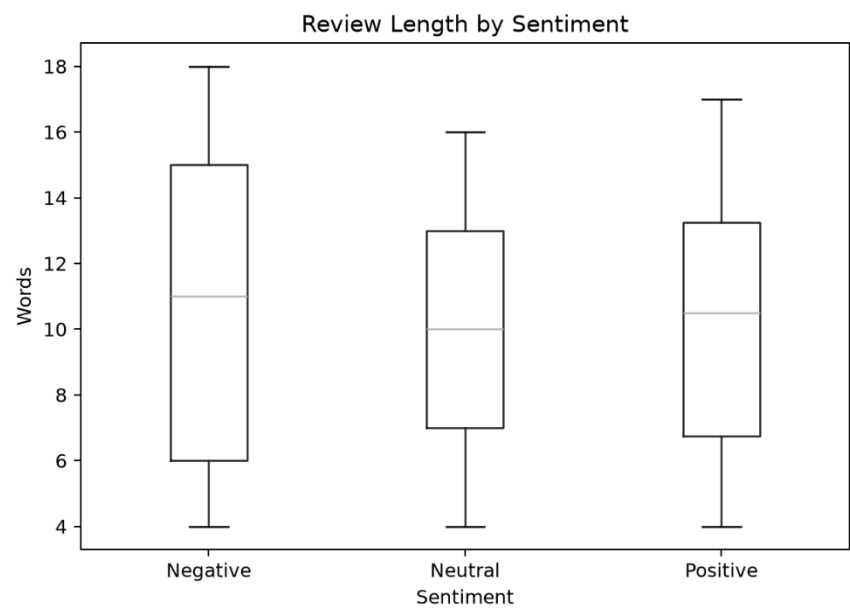


Figure 2: Review Length by Sentiment

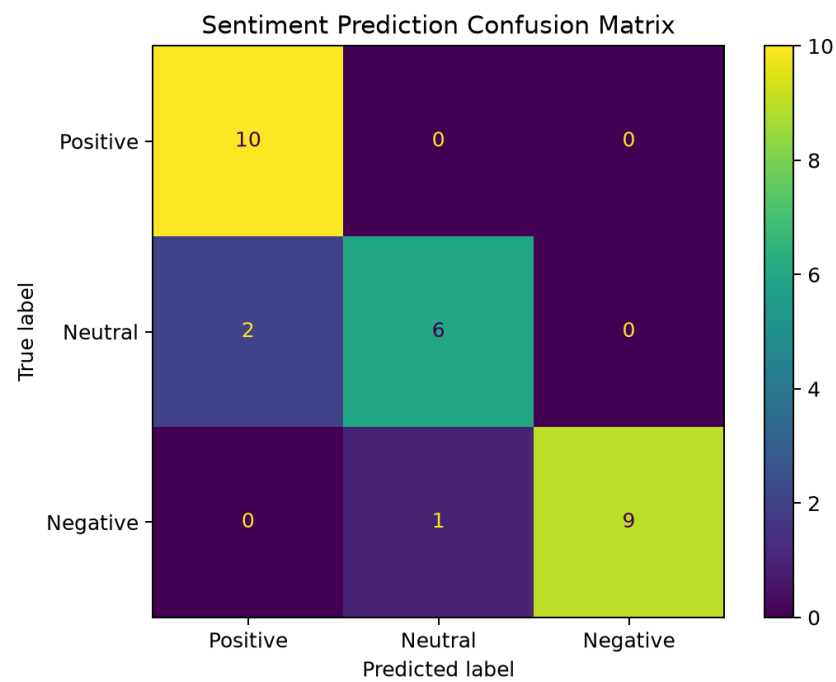


Figure 3: Sentiment Prediction Confusion Matrix

10. VISUALIZATION SUMMARY

- The sentiment distribution chart shows the relative number of Positive, Neutral, and Negative records in the labeled dataset.
- The review-length visualization provides an additional view of the textual data across sentiment categories.
- The confusion matrix helps evaluate how correctly the model classified the different sentiment categories.
- Together, these visualizations support easier interpretation of the sentiment analysis results.

The distribution chart contains 110 labeled feedback records, while the confusion matrix and model evaluation are based on the separate 28-record test set. Keeping these scopes distinct avoids mixing descriptive dataset totals with model-test results.

11. SENTIMENT PREDICTION MODEL

A machine-learning classification model was developed to predict the sentiment of customer feedback. The project uses TF-IDF (Term Frequency-Inverse Document Frequency) for converting textual feedback into numerical features. The classification algorithm used is Logistic Regression.

Model workflow

1. Clean customer feedback.
2. Split the dataset into training and testing data.
3. Convert text into TF-IDF features.
4. Train Logistic Regression.
5. Predict sentiment for the test data.
6. Compare predicted and actual sentiment.
7. Evaluate model performance.

11.1 MODEL EVALUATION

The model was evaluated using the test dataset.

Metric	Result
Test set size	28
Accuracy	0.8929
Accuracy percentage	89.29%

Classification Report

Sentiment	Precision	Recall	F1-Score	Support
Negative	1.00	0.90	0.95	10
Neutral	0.86	0.75	0.80	8
Positive	0.83	1.00	0.91	10
Accuracy	-	-	0.89	28
Macro Average	0.90	0.88	0.89	28
Weighted Average	0.90	0.89	0.89	28

The model achieved an accuracy of 89.29% on the test set.

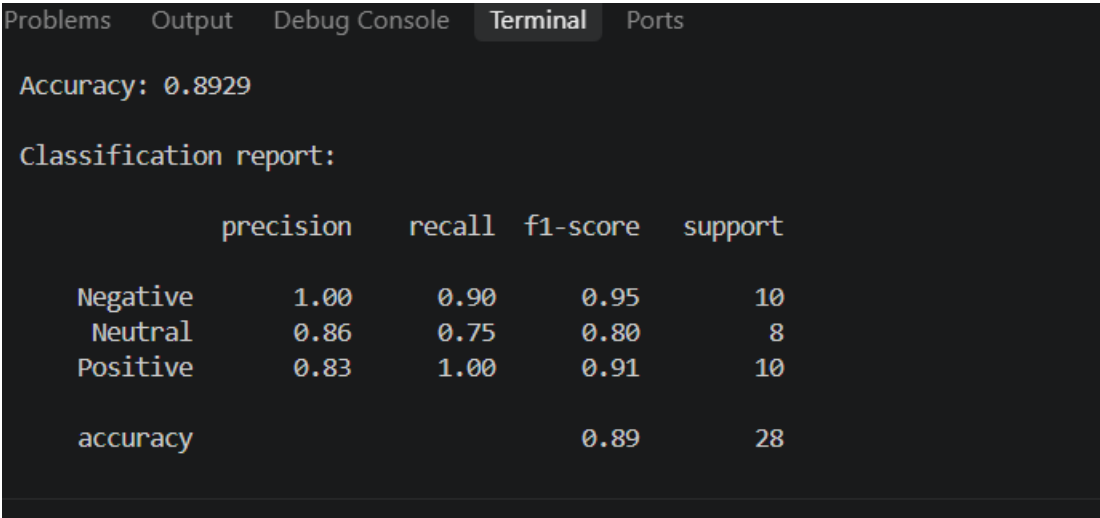


Figure 4: Python Terminal - Sentiment Model Evaluation Output

12. POWER BI DASHBOARD

An interactive dashboard was created using Microsoft Power BI to present the customer sentiment results.

- Total Test Feedback
- Predicted Customer Sentiment column chart
- Sentiment Distribution donut chart
- Customer Feedback & Sentiment Prediction table

Dashboard Component	Value / Purpose
Total Test Feedback	28
Positive predictions	12
Negative predictions	9
Neutral predictions	7
Sentiment visualization	Column chart and donut chart
Feedback review	Table of feedback, actual sentiment and predicted sentiment

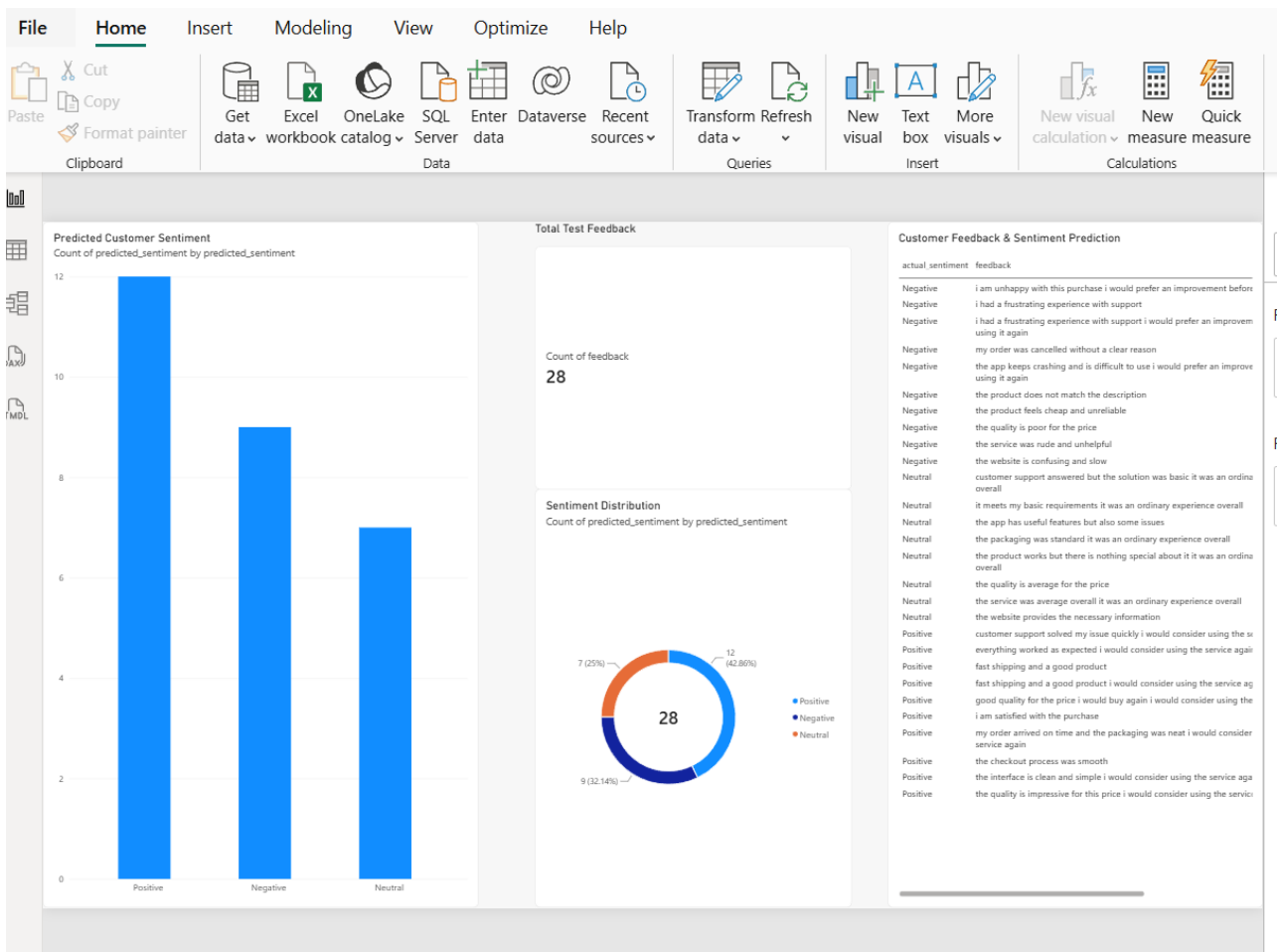


Figure 5: Power BI Customer Feedback Sentiment Dashboard

13. RESULTS AND FINDINGS

The Customer Feedback Sentiment Analysis project successfully processed customer feedback and classified the test feedback into Positive, Neutral, and Negative sentiment categories.

- Total test feedback: 28
- Positive predictions: 12
- Negative predictions: 9
- Neutral predictions: 7
- Model accuracy: 89.29%
- Model: TF-IDF + Logistic Regression

The confusion matrix shows the relationship between actual and predicted sentiment categories. The Power BI dashboard provides an interactive visual representation of the sentiment results and individual customer feedback.

14. CONCLUSION

The Customer Feedback Sentiment Analysis project demonstrates how data science and machine-learning techniques can be used to analyze textual customer feedback.

Python was used for data preparation, preprocessing, visualization, and sentiment classification. TF-IDF was used to convert text into numerical features, and Logistic Regression was used to predict sentiment.

The model achieved an accuracy of 89.29% on the test dataset. Power BI was used to create an interactive dashboard that presents the sentiment distribution and prediction results in an understandable format.

15. FUTURE ENHANCEMENTS

- Use a larger real-world customer feedback dataset.
- Include feedback from multiple products or services.
- Improve preprocessing using advanced natural-language processing techniques.
- Test additional classification algorithms.
- Apply deep-learning or transformer-based NLP models.
- Add sentiment confidence scores.
- Add interactive Power BI filters.
- Include customer category or product-level analysis.
- Add real-time feedback analysis.
- Deploy the sentiment analysis system as a web application.

16. PROJECT SCREENSHOTS

The following screenshot provides evidence of the project structure in Visual Studio Code, including the dataset, documentation, generated outputs, source code, and Power BI report file.

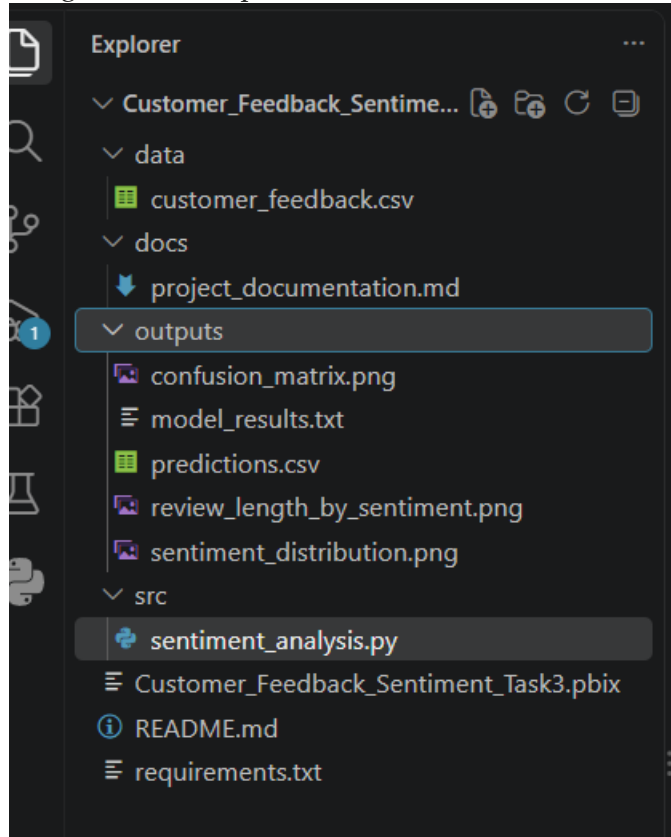


Figure 6: Customer Feedback Sentiment project structure in Visual Studio Code

The organized folder structure supports the CODTECH requirement to maintain the complete project and supporting files in a clear repository structure.

Folder / File	Purpose
data	Dataset
docs	Documentation
outputs	Charts, predictions and model results
src	Python implementation
README.md	Project information and scope
requirements.txt	Python dependencies
Customer_Feedback_Sentiment_Task3.pbix	Power BI dashboard

16. PROJECT SCREENSHOTS (CONTINUED)

```

11
12 ROOT = Path(__file__).resolve().parents[1]
13 DATA = ROOT / "data" / "customer_feedback.csv"
14 OUT = ROOT / "outputs"
15 OUT.mkdir(exist_ok=True)
16
17 df = pd.read_csv(DATA)
18
19 def clean_text(text: str) -> str:
20     text = str(text).lower()
21     text = re.sub(r"http\S+|www\S+", " ", text)
22     text = re.sub(r"^[^a-z\s]", " ", text)
23     text = re.sub(r"\s+", " ", text).strip()
24     return text
25
26 df["clean_feedback"] = df["feedback"].apply(clean_text)
27
28 # 1) Sentiment distribution
29 counts = df["sentiment"].value_counts().reindex(["Positive", "Neutral", "Negative"]).fillna(0)
30 ax = counts.plot(kind="bar", title="Customer Feedback Sentiment Distribution")
31 ax.set_xlabel("Sentiment")
32 ax.set_ylabel("Number of Reviews")
33 plt.tight_layout()

```

Figure 7: Python Source Code - Data Loading and Text Preprocessing

This source-code section demonstrates project-path setup, dataset loading using Pandas, output-folder setup, the text-cleaning function, and creation of the cleaned feedback field.

16. PROJECT SCREENSHOTS (CONTINUED)

```

37
38 # 2) Review text length by sentiment
39 df["text_length"] = df["clean_feedback"].str.split().str.len()
40 ax = df.boxplot(column="text_length", by="sentiment", grid=False)
41 plt.suptitle("")
42 ax.set_title("Review Length by Sentiment")
43 ax.set_xlabel("Sentiment")
44 ax.set_ylabel("Words")
45 plt.tight_layout()
46 plt.savefig(OUT / "review_length_by_sentiment.png", dpi=180)
47 plt.show()
48 plt.close()
49
50 # 3) Train a sentiment prediction model
51 X_train, X_test, y_train, y_test = train_test_split(
52     df["clean_feedback"],
53     df["sentiment"],
54     test_size=0.25,
55     random_state=42,
56     stratify=df["sentiment"]
57 )
58

```

Figure 8: Python Source Code - Training Data Preparation

```

58
59 model = Pipeline([
60     ("tfidf", TfidfVectorizer(stop_words="english", ngram_range=(1, 2))),
61     ("classifier", LogisticRegression(max_iter=1000))
62 ])
63
64 model.fit(X_train, y_train)
65 pred = model.predict(X_test)
66
67 accuracy = accuracy_score(y_test, pred)
68 print(f"Accuracy: {accuracy:.4f}")
69 print("\nClassification report:\n")
70 print(classification_report(y_test, pred))
71
72 # 4) Confusion matrix
73 cm = confusion_matrix(y_test, pred, labels=["Positive", "Neutral", "Negative"])
74 disp = ConfusionMatrixDisplay(
75     confusion_matrix=cm,
76     display_labels=["Positive", "Neutral", "Negative"]
77 )
78 disp.plot()
79 plt.title("Sentiment Prediction Confusion Matrix")
80 plt.tight_layout()

```

Figure 9: Python Source Code - TF-IDF, Logistic Regression and Evaluation

16. PROJECT OUTPUT EVIDENCE

Negative	1.00	0.90	0.95	10
Neutral	0.86	0.75	0.80	8
Positive	0.83	1.00	0.91	10
accuracy			0.89	28
macro avg	0.90	0.88	0.89	28
weighted avg	0.90	0.89	0.89	28

PS C:\Users\ragha\Downloads\Customer_Feedback_Sentiment_CODTECH_T

Figure 10: Python Terminal - Classification Report and Final Output

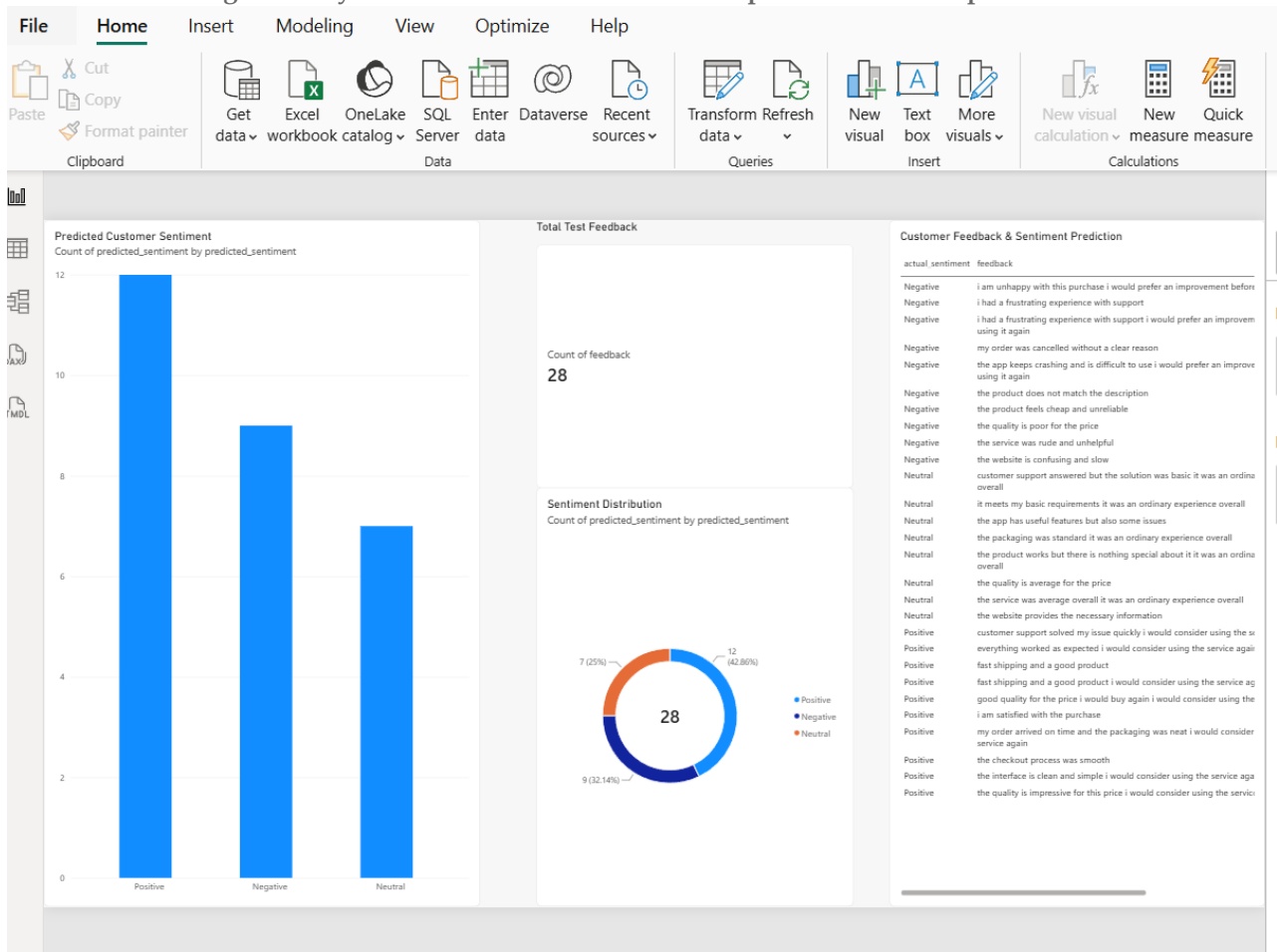


Figure 11: Power BI Dashboard Output

17. SOURCE CODE

The Customer Feedback Sentiment Analysis project was implemented using Python in Visual Studio Code. Pandas was used for dataset loading and data manipulation. Text preprocessing was performed to clean the customer feedback. TF-IDF was used for text feature extraction, while Logistic Regression was used for sentiment classification. Matplotlib was used for data visualization, and Power BI was used to create the interactive dashboard.

Main implementation stages

- Import statements and dataset loading
- Text cleaning and preprocessing
- Sentiment analysis
- Data visualization
- TF-IDF feature extraction
- Logistic Regression model training
- Prediction and evaluation
- Confusion matrix generation
- Power BI dashboard creation

Project Item	Included
Dataset	customer_feedback.csv
Source code	src/sentiment_analysis.py
Documentation	docs/project_documentation.md
Outputs	Charts, predictions.csv and model_results.txt
Dashboard	Customer_Feedback_Sentiment_Task3.pbix
Academic report	Customer Feedback Sentiment Analysis - MCA Project Report

The complete source code and project artifacts are maintained in the GitHub repository for submission and reference.

END OF PROJECT REPORT